MTH 331(A) Dr. James Ashe
10.1 Worksheet Friday 6^{th} September, 2013 **Name:**

Definition. Let f be a function defined at a. The n^{th} *order Taylor polynomial* for f with its *center* at a, denoted p_n is

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

1. Complete the following.

a. Find a Taylor polynomial of order $n = 3$ for $\ln(x)$.

b. Use the Taylor polynomial from above to to approximate $\ln(1.07)$.

c. Find the absolute error of the approximation above.

2. Use a Taylor polynomial of order $n = 5$ to approximate $\cos(0.1)$.

3. Approximate $\dfrac{1}{\sqrt{4.08}}$ using a Taylor polynomial of order $n = 2$.